

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

8875/02

Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **any one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	13
2	10
3	8
4	9
Section B	
5 / 6	20
Total	60
Percentage	
Grade	

This document consists of **12** printed pages.



Innova Junior College

[Turn over

Section A

Answer **all** questions.

- 1** Blackflies are aggressive blood-feeders on people, domestic livestock and a variety of warm-blooded wildlife. Its larvae live in rivers and feed by filtering particles out of the water.

A method of controlling blackflies is to add particles of an inactive proteinous toxin to the water upstream of where the larvae are feeding. The toxin will be activated by a protease secreted by the stomach of the larva.

The active toxin then binds to a receptor molecule on the membranes of the gut epithelial cell. This damages the epithelial cells and kills the larva.

Fig. 1.1 is an electron micrograph of a stomach cell of the blackfly larva.

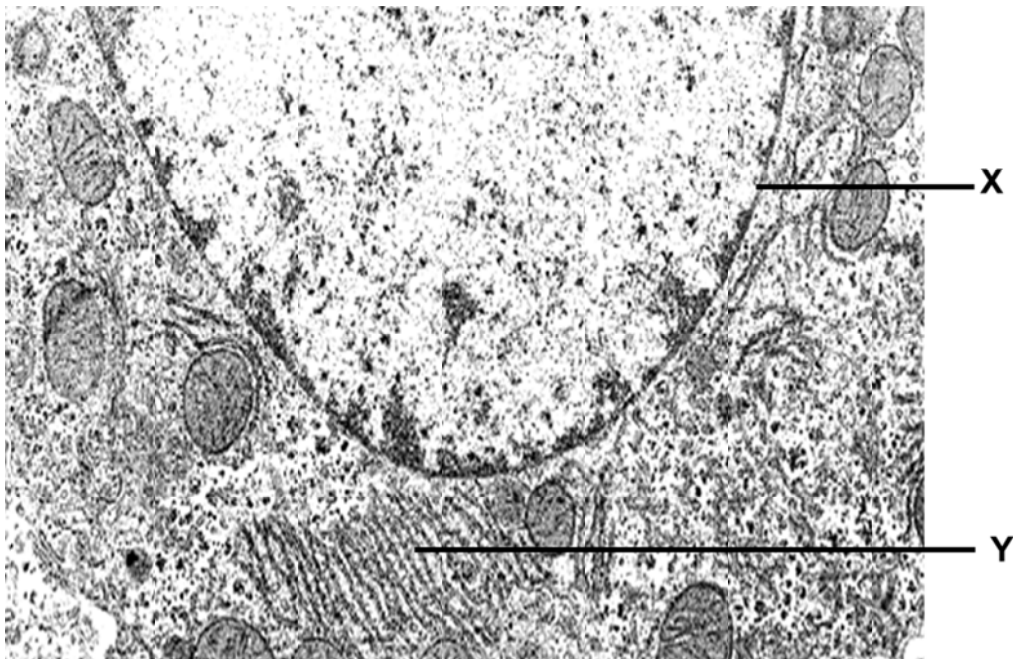


Fig. 1.1

- (a) (i)** Describe how organelle X is adapted for genetic expression.

.....

.....

.....

..... [2]

- (ii) Describe how insulin, after being synthesized by ribosomes on Y, is exported out of the cell.

.....

.....

.....

.....

.....

..... [4]

Fig. 1.2 shows the results of an experiment to determine the optimum pH for activation of the toxin by the stomach protease.

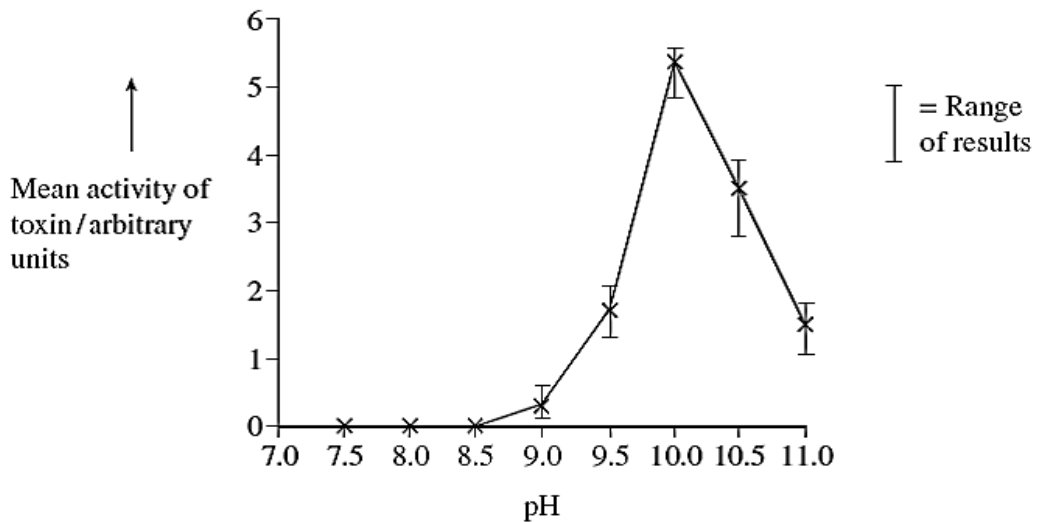


Fig. 1.2

- (b) With reference to Fig. 1.2, describe and explain the effect of pH on the mean activity of the toxin.

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.....

.....

..... [3]

- (c) Suggest how the protease in the larva's digestive system activates the toxin.

.....

 [2]

- (d) The use of this toxin has been shown to be effective in eliminating the blackfly larvae population while having no adverse effect on other organisms living in the river.

Suggest **two** reasons why the toxin can be used for selective biological control of the blackfly.

.....

 [2]

[Total: 13]

- 2 Fig. 2.1 shows the light dependent reactions occurring in the chloroplast.

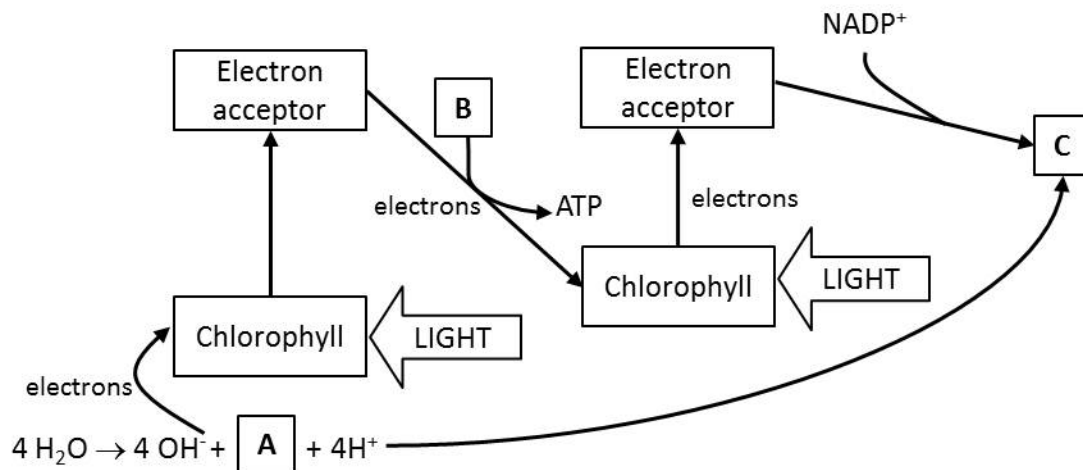


Fig. 2.1

- (a) (i) Name the compounds A – C.

A
 B
 C

[1]

- (ii) Explain the role of phospholipids in the synthesis of ATP.

----- [2]

- (b) With reference to Fig. 2.1, explain the role of water in photosynthesis.

----- [2]

Scientists investigated the effects of temperature on the rate of respiration and photosynthesis in creeping azalea. Fig. 2.2 shows the results of temperature on the net rate of photosynthesis at three different light intensities.

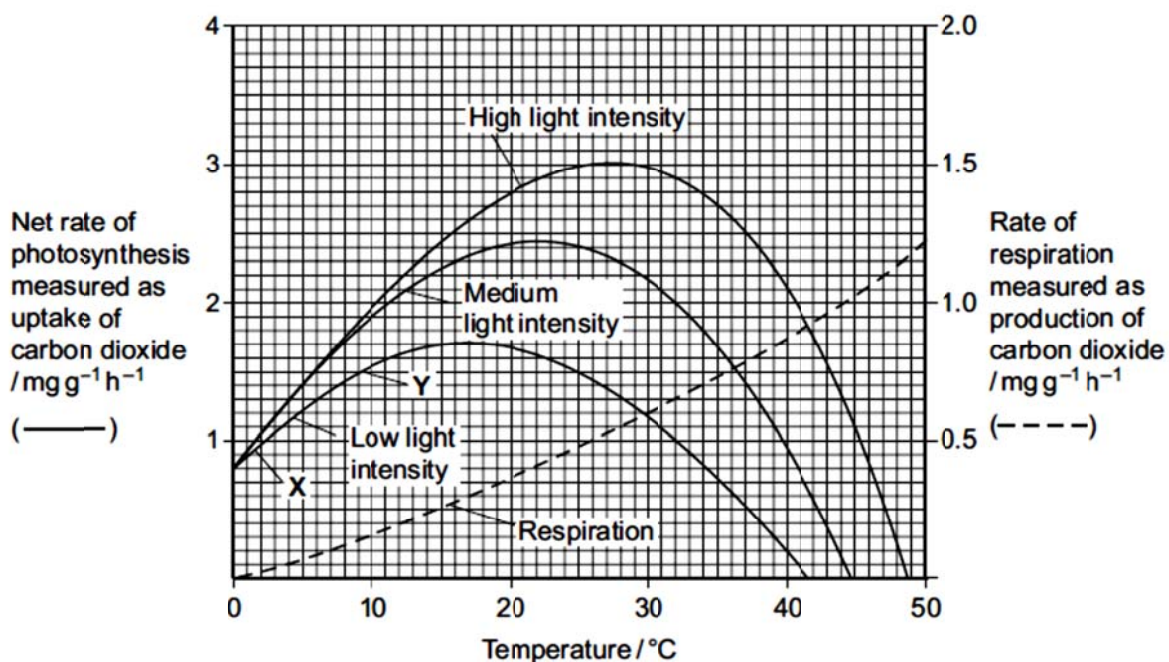


Fig. 2.2

- (c) (i) Name the factors that limited the rate of photosynthesis between X and Y.

----- [1]

(ii) Explain your answers in **c(i)**.

----- [2]

- (d) Creeping azalea is a plant which grows on mountains. Scientists predict that in the area where this plant grows the mean summer temperature is likely to rise from 20°C to 23°C. It is also likely to become much cloudier.

Suggest how these changes are likely to affect the growth of creeping azalea.

----- [2]

[Total: 10]

- 3 The Rhesus blood group is genetically controlled. The gene for the Rhesus blood group is found on chromosome 1 and has two alleles. The allele for Rhesus positive, **R**, is dominant to that for Rhesus negative, **r**.

Another gene coding for the ABO blood groups is found on chromosome 19 and both genes play an important role in the determination of suitable donors for blood transfusion.

Fig. 3 shows the inheritance of the Rhesus blood group in one family.

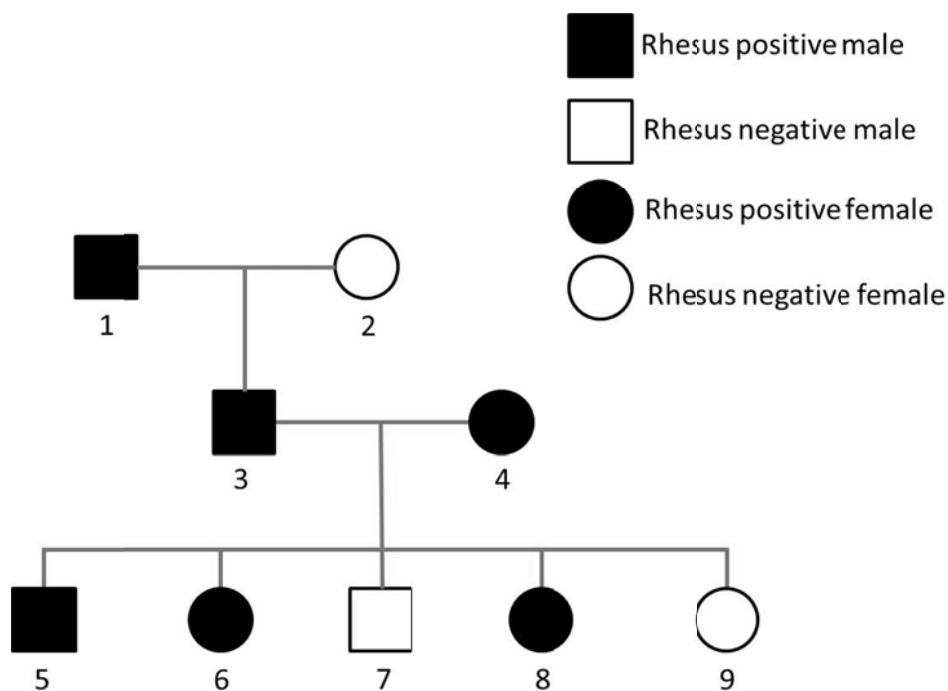


Fig. 3

- (a) Explain the term *dominant*.

.....

.....

.....

..... [2]

- (b) State **one** piece of evidence from Fig. 3 that shows **R** is the dominant allele in the determination of the Rhesus phenotype.

.....

..... [1]

- (c) Individual 9 with blood group B married a male with blood group A who is Rhesus positive. The couple had a daughter who is Rhesus positive with the blood group AB and a son who is Rhesus negative with the blood group O.

Using appropriate symbols, illustrate the information in a genetic diagram.

[4]

- (d) State the probability that the couple's next child will be a boy who is Rhesus negative with the blood group A.

Probability : [1]

[Total: 8]

4 Plasmids are often used as vectors in genetic engineering.

(a) (i) Explain what is meant by a *vector*.

----- [1]

(ii) Describe the role of restriction endonucleases in the formation of recombinant plasmids.

----- [2]

- (b) Fig. 4.1 shows the position of four unique restriction sites, J, K, L and M on a plasmid.

Gel electrophoresis is carried out to separate the restriction fragments after treatment with endonucleases. The results of electrophoresis are shown in Fig. 4.2.

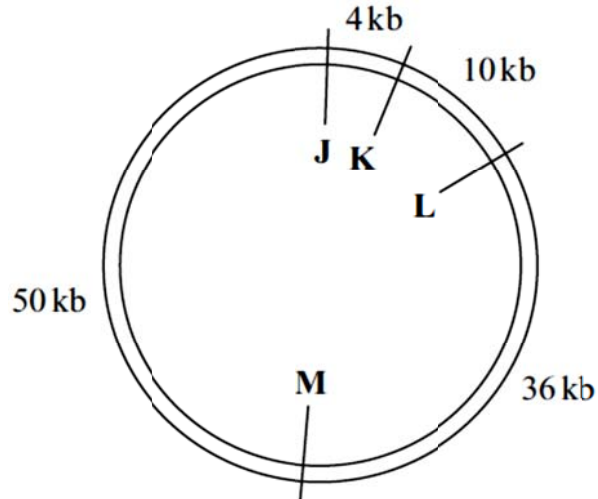


Fig. 4.1

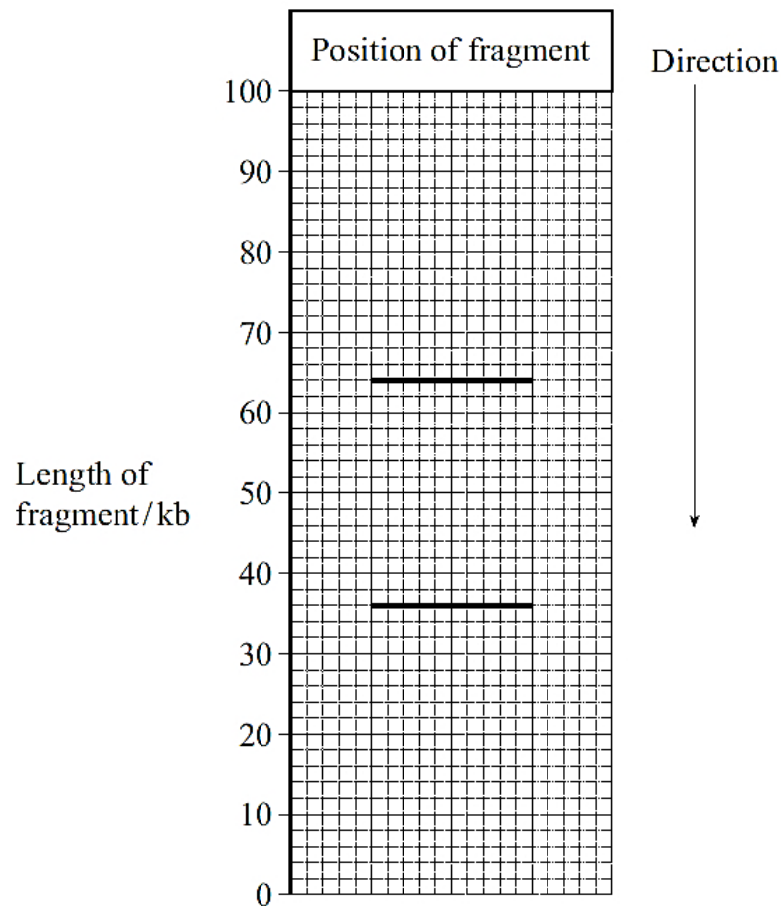


Fig. 4.2

With reference to Fig. 4.1 and 4.2,

- (i) state which restriction sites were cleaved.

..... [1]

- (ii) Explain how the restriction fragments were separated by gel electrophoresis.

..... [3]

- (iii) Restriction site M is found within the *lac Z* gene and restriction site L is found within the tetracyclin resistant gene.

Suggest why restriction enzyme L is unsuitable to be used in the formation of a recombinant plasmid.

..... [2]

[Total: 9]

Section B
Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 5** **(a)** Describe the neo-Darwinism theory of evolution. [6]
- (b)** Explain with relevant examples, how anatomical, embryological and molecular homology supports natural selection. [8]
- (c)** Explain the ways in which islands favour speciation. [6]
- [Total: 20]
- 6** **(a)** Explain the normal functions of stem cells in a living organism. [6]
- (b)** Discuss the goals and implications of the human genome project, including the benefits and difficult ethical concerns for humans. [8]
- (c)** Comment on the statement “GM crops can make a relevant contribution to solving health problems in developing countries.” [6]
- [Total: 20]